

OS Lab

Skill-9

To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation.

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

#define MAX_INPUT 1024

int main()
{
    char input[MAX_INPUT];
    char current_dir[PATH_MAX];
    char previous_dir[PATH_MAX];

    /* Store the starting directory */
    if (getcwd(current_dir, sizeof(current_dir)) == NULL)
    {
        perror("getcwd");
        return 1;
    }

    strcpy(previous_dir, current_dir);

    printf("Directory Navigation Shell\n");
    printf("Commands:\n");
    printf(" cd <directory> - Change directory\n");
    printf(" cd -          - Go to previous directory\n");
    printf(" pwd          - Show current directory\n");
    printf(" exit         - Exit\n\n");

    while (1)
    {
        printf("nav-shell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
```

```

    break;

input[strcspn(input, "\n")] = '\0';

if (strcmp(input, "exit") == 0)
    break;

/* Show current directory */
if (strcmp(input, "pwd") == 0)
{
    if (getcwd(current_dir, sizeof(current_dir)) != NULL)
        printf("%s\n", current_dir);

    continue;
}

/* Handle cd */
if (strncmp(input, "cd", 2) == 0)
{
    char *path = input + 2;

    while (*path == ' ')
        path++;

    /* cd without argument */
    if (*path == '\0')
    {
        printf("cd: missing directory\n");
        continue;
    }

    /* Save current directory before changing */
    if (getcwd(current_dir, sizeof(current_dir)) == NULL)
    {
        perror("getcwd");
        continue;
    }

    /* Handle cd - */
    if (strcmp(path, "-") == 0)
    {
        if (chdir(previous_dir) != 0)
        {
            perror("cd");
        }
    }
}

```

```

        continue;
    }

    strcpy(current_dir, previous_dir);

    if (getcwd(previous_dir, sizeof(previous_dir)) == NULL)
    {
        perror("getcwd");
        continue;
    }

    printf("%s\n", current_dir);
    continue;
}

/* Change to requested directory */
if (chdir(path) != 0)
{
    perror("cd");
    continue;
}

/* Update previous directory */
strcpy(previous_dir, current_dir);

/* Get new current directory */
if (getcwd(current_dir, sizeof(current_dir)) != NULL)
{
    printf("Changed directory to: %s\n", current_dir);
}

continue;
}

printf("Invalid command: %s\n", input);
}

return 0;
}

```

```

lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ cd ~/OSLab/shell_lab
nano cd_navigation.c
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ gcc cd_navigation.c -o cd_navigation
cd_navigation.c:7: warning: "MAX_INPUT" redefined
   7 | #define MAX_INPUT 1024
     |
In file included from /usr/include/x86_64-linux-gnu/bits/local_lim.h:38,
                 from /usr/include/x86_64-linux-gnu/bits/posix1_lim.h:161,
                 from /usr/include/limits.h:195,
                 from /usr/lib/gcc/x86_64-linux-gnu/13/include/limits.h:205,
                 from /usr/lib/gcc/x86_64-linux-gnu/13/include/syslimits.h:7,
                 from /usr/lib/gcc/x86_64-linux-gnu/13/include/limits.h:34,
                 from cd_navigation.c:5:
/usr/include/linux/limits.h:11: note: this is the location of the previous definition
   11 | #define MAX_INPUT      255      /* size of the type-ahead buffer */
     |
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ./cd_navigation
Directory Navigation Shell
Commands:
  cd <directory> - Change directory
  cd -          - Go to previous directory
  pwd           - Show current directory
  exit         - Exit

nav-shell> pwd
/home/lahari/OSLab/shell_lab
nav-shell> cd ..
Changed directory to: /home/lahari/OSLab
nav-shell> pwd
/home/lahari/OSLab
nav-shell> cd -
/home/lahari/OSLab/shell_lab
nav-shell> pwd

```

```
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ./cd_navigation
Directory Navigation Shell
Commands:
  cd <directory>  - Change directory
  cd -            - Go to previous directory
  pwd            - Show current directory
  exit           - Exit

nav-shell> pwd
/home/lahari/OSLab/shell_lab
nav-shell> cd ..
Changed directory to: /home/lahari/OSLab
nav-shell> pwd
/home/lahari/OSLab
nav-shell> cd -
/home/lahari/OSLab/shell_lab
nav-shell> pwd
/home/lahari/OSLab/shell_lab
nav-shell> cd /tmp
Changed directory to: /tmp
nav-shell> pwd
/tmp
nav-shell> cd -
/home/lahari/OSLab/shell_lab
nav-shell> exit
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ |
```